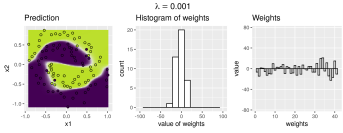
# Introduction to Machine Learning

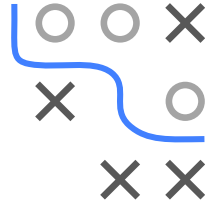
# Regularization in Non-Linear Models and Structural Risk Minimization




## Learning goals

- Understand that regularization and parameter shrinkage can be applied to non-linear models
- Know structural risk minimization

# SUMMARY: REGULARIZED RISK MINIMIZATION



If we should define (supervised) ML in only one line, this might be it:

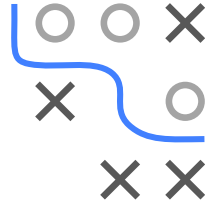
$$\min_{\boldsymbol{\theta}} \mathcal{R}_{\text{reg}}(\boldsymbol{\theta}) = \min_{\boldsymbol{\theta}} \left( \sum_{i=1}^{n} L\left(y^{(i)}, f\left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}\right)\right) + \lambda \cdot J(\boldsymbol{\theta}) \right)$$

We can choose for a task at hand:

- the **hypothesis space** of $f$, which determines how features can influence the predicted $y$
- the **loss** function $L$, which measures how errors should be treated
- the **regularization** $J(\boldsymbol{\theta})$, which encodes our inductive bias and preference for certain simpler models

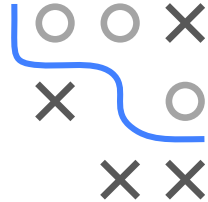
By varying these choices one can construct a huge number of different ML models. Many ML models follow this construction principle or can be interpreted through the lens of regularized risk minimization.

# REGULARIZATION IN NONLINEAR MODELS

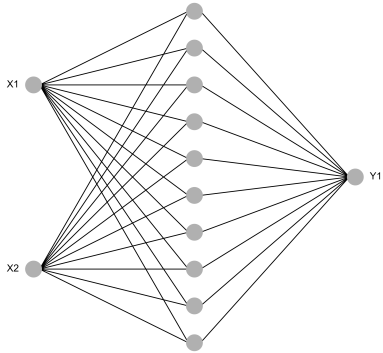


- So far we have mainly considered regularization in LMs.
- Can also be applied to non-linear models (with numeric parameters), where it is often important to prevent overfitting.
- Often, non-linear models can be seen as LMs based on internally transformed features.
- Here, we typically use $L2$ regularization, which still results in parameter shrinkage and weight decay.
- Adding regularization is commonplace and sometimes crucial in non-linear methods such as NNs, SVMs, or boosting.
- By adding regularization, prediction surfaces in regression and classification become smoother.

# REGULARIZATION IN NONLINEAR MODELS



**Setting**: Classification for the $\mathrm{spirals}$ data. Neural network with single hidden layer containing 10 neurons, regularized with $L2$.
Varying $\lambda$ affects smoothness of the decision boundary and magnitude of network weights:



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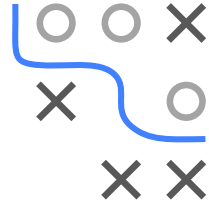

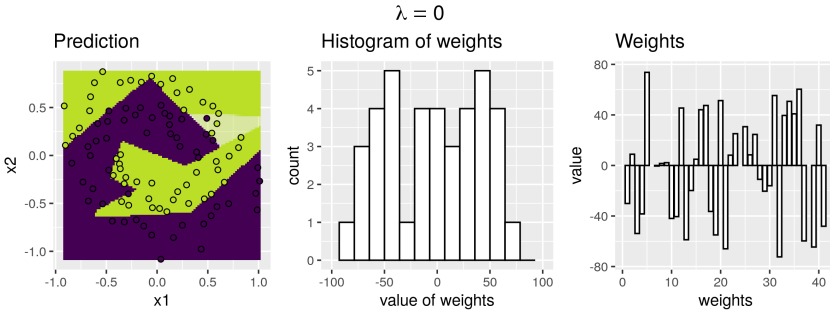

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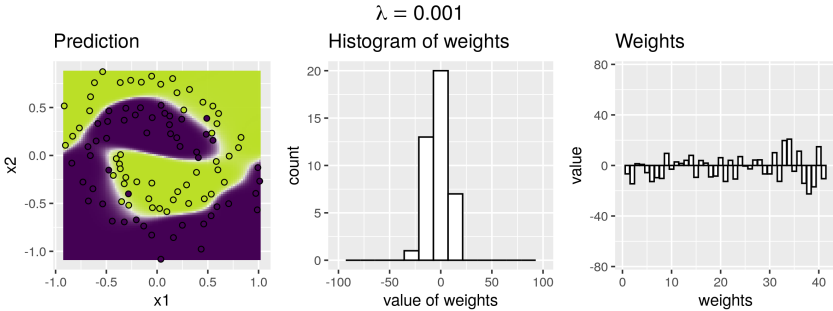

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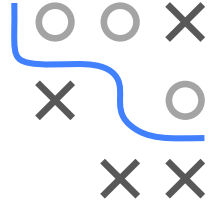

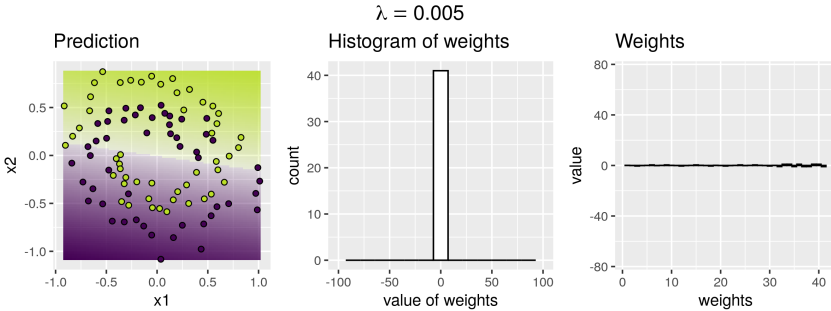

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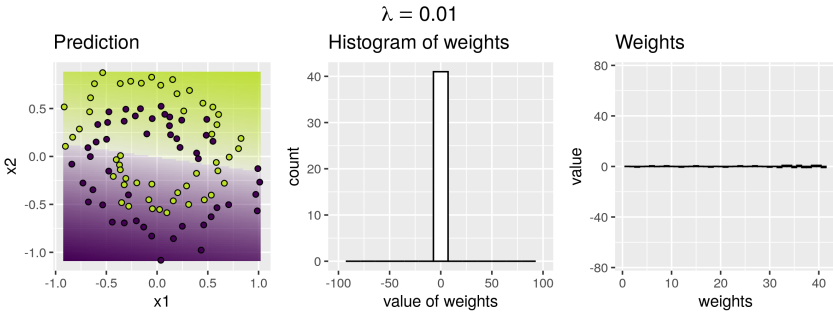
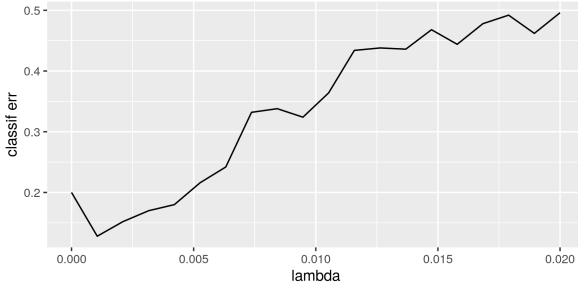


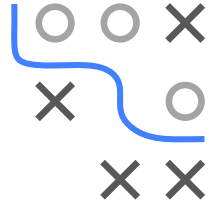
# REGULARIZATION IN NONLINEAR MODELS

The prevention of overfitting can also be seen in CV. Same settings as before, but each $\lambda$ is evaluated with repeated CV (10 folds, 5 reps).

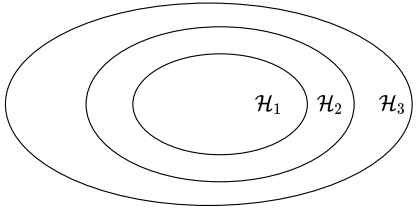



We see the typical U-shape with the sweet spot between overfitting (LHS, low $\lambda$) and underfitting (RHS, high $\lambda$) in the middle.

# STRUCTURAL RISK MINIMIZATION

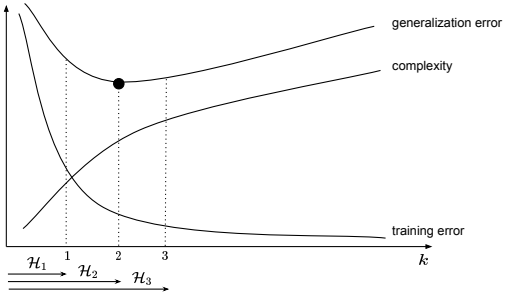


- Thus far, we only considered adding a complexity penalty to empirical risk minimization.
- Instead, structural risk minimization (SRM) assumes that the hypothesis space $\mathcal{H}$ can be decomposed into increasingly complex hypotheses (size or capacity): $\mathcal{H} = \cup_{k \geq 1} \mathcal{H}_k$.
- Complexity parameters can be the, e.g. the degree of polynomials in linear models or the size of hidden layers in neural networks.



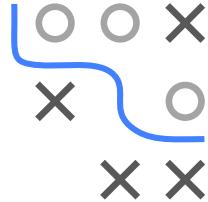
# STRUCTURAL RISK MINIMIZATION / 2

- SRM chooses the smallest $k$ such that the optimal model from $\mathcal{H}_k$ found by ERM or RRM cannot significantly be outperformed by a model from a $\mathcal{H}_m$ with $m > k$.
- By this, the simplest model can be chosen, which minimizes the generalization bound.
- One challenge might be choosing an adequate complexity measure, as for some models, multiple complexity measures exist.

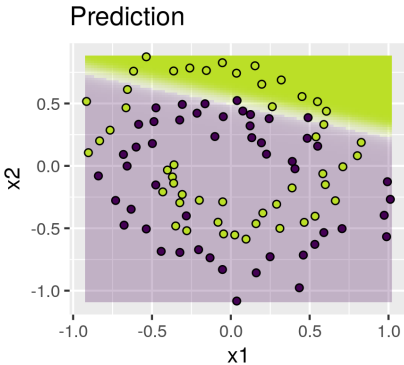


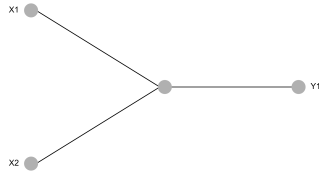
# STRUCTURAL RISK MINIMIZATION

**Setting**: Classification for the $\mathrm{spirals}$ data. NN with 1 hidden layer, and fixed (small) L2 penalty.
Varying the size of the hidden layer affects smoothness of the decision boundary:



size of hidden layer = 1

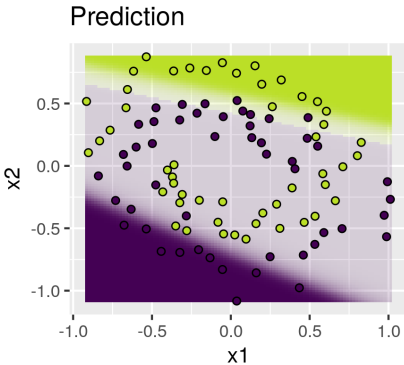


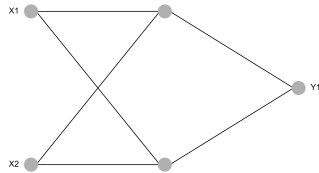



# STRUCTURAL RISK MINIMIZATION

**Setting**: Classification for the $\mathrm{spirals}$ data. NN with 1 hidden layer, and fixed (small) L2 penalty.
Varying the size of the hidden layer affects smoothness of the decision boundary:

$$\text{size of hidden layer} = 2$$

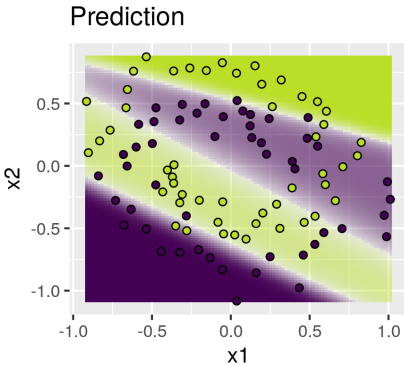


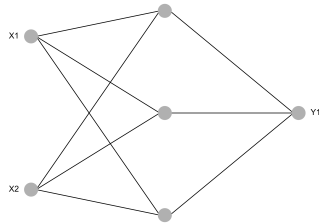



# STRUCTURAL RISK MINIMIZATION

**Setting**: Classification for the $\mathrm{spirals}$ data. NN with 1 hidden layer, and fixed (small) L2 penalty.
Varying the size of the hidden layer affects smoothness of the decision boundary:

$$\text{size of hidden layer} = 3$$

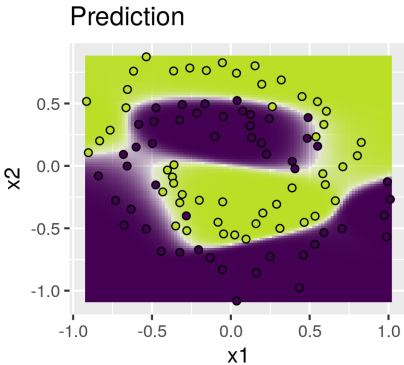


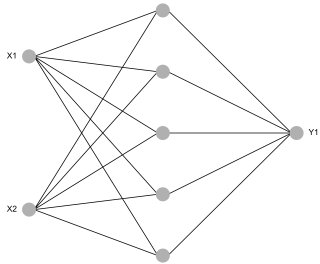



# STRUCTURAL RISK MINIMIZATION

**Setting**: Classification for the $\mathrm{spirals}$ data. NN with 1 hidden layer, and fixed (small) L2 penalty.
Varying the size of the hidden layer affects smoothness of the decision boundary:

$$\text{size of hidden layer} = 5$$



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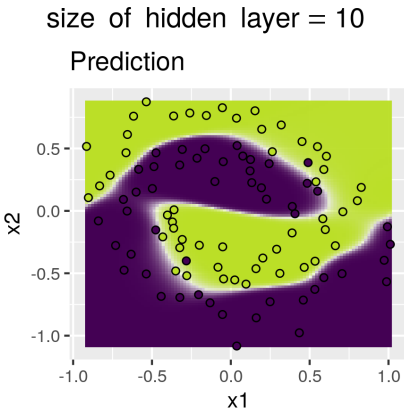


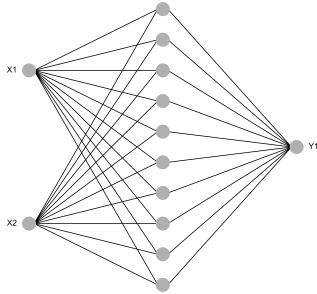

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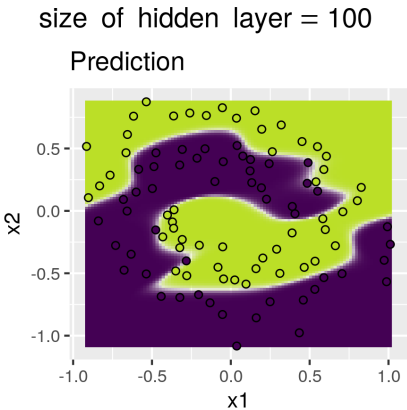
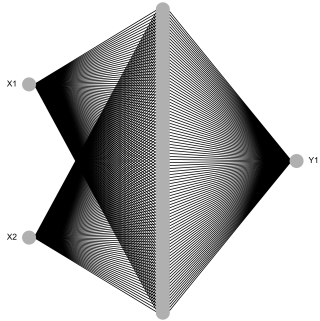
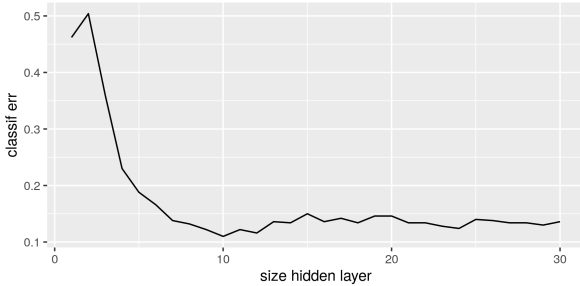


# STRUCTURAL RISK MINIMIZATION

Again, complexity vs CV score.



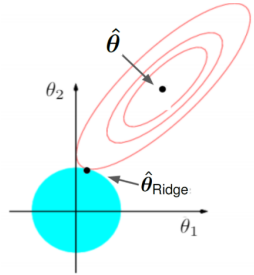

A minimal model with good generalization seems to have ca. 6-8 hidden neurons.

# STRUCTURAL RISK MINIMIZATION AND RRM

Note that normal RRM can also be interpreted through SRM, if we rewrite the penalized ERM as constrained ERM.

$$\min_{\boldsymbol{\theta}} \quad \sum_{i=1}^{n} L\left(y^{(i)}, f\left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}\right)\right)$$

$$\text{s.t.} \quad \|\boldsymbol{\theta}\|_2^2 \leq t$$




We can interpret going through $\lambda$ from large to small as through $t$ from small to large. This constructs a series of ERM problems with hypothesis spaces $\mathcal{H}_\lambda$, where we constrain the norm of $\boldsymbol{\theta}$ to unit balls of growing size.